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# Pregnancy Establishment and Diagnosis in Livestock

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## Keywords

pregnancy, livestock, reproduction, gestation, embryo, agriculture

## Abstract

This comprehensive review explores the complex processes of reproduction, pregnancy establishment, and pregnancy diagnostic methods in cattle, sheep, goats, swine, horses, and camelids. It provides an overview of the history of pregnancy detection and an in-depth exploration of the physiology of pregnancy in livestock. The detection of conceptus tissue and fluids, conceptus-produced hormones, and maternal responses to conceptus signals, crucial for pregnancy diagnosis, are also discussed in detail, as are emerging methods for pregnancy diagnosis in livestock species. Overall, this review emphasizes the direct impact of pregnancy diagnosis and efficient pregnancy management for profitability of livestock enterprises.



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# INTRODUCTION

Ensuring a steady supply of offspring is crucial for herd/flock maintenance, growth, and production of livestock-derived food and nonfood products, as well as for work, sport, and companionship. This involves timely insemination, successful establishment of pregnancy, and accurate pregnancy testing. Livestock species in this review include those that are either polyestrous (cattle, pigs) or seasonally polyestrous (horses, sheep, certain camelids), emphasizing the need for optimization of breeding management. Moreover, because a significant proportion of inseminations do not result in a pregnancy carried to term, early timing and accuracy in pregnancy diagnosis are vital for informed decision-making by livestock producers. This includes implementing rebreeding protocols, veterinary treatment, or animal culling. The choice of pregnancy diagnosis method varies among livestock species due to differences in reproductive physiology, production systems, and production costs. Thus, this review explores the current knowledge of pregnancy establishment and diagnosis in key livestock species including cattle, sheep, goat, swine, horses, and camelids.

## History of Pregnancy Diagnosis in Livestock

Readers interested in the history of pregnancy detection methods from the Pharaonic (Egyptian) period to the twentieth century should read the engaging reviews by Baylon (1) and Cowie (2). Historically, the most common method of pregnancy detection has been the observation of lack of return to estrus (termed anestrus) after breeding. However, this approach is inaccurate, because factors other than pregnancy can also cause a female to be anestrus, and estrus behaviors can be ephemeral and easily missed. Later in gestation, fetal-placental growth and fluid accumulation in placental membranes provide opportunities for external assessment or manual palpation for pregnancy confirmation. For instance, a method called ballottement, used in sheep and goats, effectively diagnoses pregnancy after approximately 90 days of gestation. In this approach, the animal is placed in a seated position and leaned back against the handler's legs; the handler then places their hands low on the abdomen and pushes on one side, feeling for the movement of the uterine contents bumping into the opposing hand. A modification of this method involves inserting a rod in the rectum and using it to move the uterine contents toward the abdominal wall for palpation.

Transrectal palpation (TRP) for pregnancy diagnosis dates to the early nineteenth century and gained widespread use by the early twentieth century (2). Despite being physically demanding and presenting some risk of injury to the animal and the practitioner, TRP of uterine contents can detect pregnancy as early as the fourth week of gestation in cattle and the third week in horses. In this approach, a trained practitioner will insert a gloved, lubricated arm into the rectum of a restrained animal and palpate the ovaries and uterus, including palpation of the amniotic vesicle, chorioallantois through the membrane slip method, placentomes, and embryo or fetus (3). Cowie (2, p. 12, emphasis in original) strongly emphasized the importance of TRP, disputing arguments stating that TRP was a "*messy, objectionable procedure*" and asserting that "*such an attitude is inexcusable and failure to carry out rectal examination for such reasons was merely incompetence*."

Other pregnancy detection methods involve the assessment of blood flow in the middle uterine and ovarian arteries. This approach uses TRP or a stethoscope inserted in the vagina to detect arterial vibration (fremitus) induced by increased blood flow (2). Except for TRP, the methods described above are ineffective until the latter third of gestation, when an increase in mammary gland size is often evident and can also be used as an indicator of impending parturition. The Benesch–Kurosawa test, involving macro- and microscopic evaluation of the vagina, is another method said to have high accuracy as early as 3 weeks of pregnancy but is apparently effective only in horses and no other species (2). Interestingly, when swine pregnancy diagnosis remained

elusive in the early twentieth century, Cowie (2, p. 19, emphasis in original) concluded that "*even toward the end of gestation, a diagnosis of probably pregnant*" was the best that could be achieved.

Pregnancy diagnosis using urine began in ancient Egypt with the observation that pregnancy urine induced seed germination. In the twentieth century, studies revealed that extracts of estrogens in pregnancy urine, initially believed to stimulate sprouting, were later attributed to a heat-labile component rather than estrogen itself (2). During this period, various tests for detecting gonadotropic hormones were developed using animals such as mice, rats, guinea pigs, rabbits, fish, frogs (genus *Rana*), and toads (genus *Xenopus*). Originally intended for women, these tests involved using urine or blood from potential pregnant individuals to stimulate follicle growth or induce ovulation/oviposition in test animals. However, their success in livestock was limited due to impracticality, cost, and time required. Nevertheless, research in mares revealed pregnancy-specific factors with gonadotrophic and estrogenic activity in the blood and urine of pregnant animals (2). Specifically, the test for gonadotropin activity achieved >90% accuracy between 45 to 110 days of mare gestation but proved ineffective in cows, sows, or ewes (4). Overall, early studies struggled to develop useful pregnancy tests from blood or urine in livestock.

## Reproductive Management in Livestock Species

Managing animal growth to achieve puberty and appropriate body size at an early age has a large impact on profitability of livestock operations. Poorly grown animals that are bred too early often exhibit suboptimal pregnancies and difficult birth, leading to complications and occasionally death. Therefore, managing livestock for efficient reproduction begins by managing pregnant females so the offspring are born at the appropriate size and then managing the offspring, so that they achieve puberty at an early age and adequate body size prior to breeding. Refer to **Table 1** for the average age at puberty for various livestock species.

**Cattle.** Dairy heifers on well-managed farms achieve puberty between 9 and 12 months of age and become eligible for insemination at 12 to 14 months of age when they reach approximately 55% to 60% of mature body weight. Considering the typical range of days to pregnancy establishment, heifers generally calve for the first time at 22 to 24 months of age. After calving, lactation begins, and cows are allowed to recover from parturition and adapt to the metabolic demands of milk production before being bred to establish a new pregnancy. During this period, referred to as the voluntary waiting period (VWP), most cows recover uterine health and return to estrous cyclicity and to a positive energy balance. The VWP in dairy cattle is typically 50 to 60 days but can be as long as 70 to 90 days in herds with high fertility (5, 6). Upon completion of the VWP, cows become eligible for pregnancy and, thus, the insemination period begins.

In most commercial dairy operations, heifers and cows are inseminated either after detection of natural estrus or after synchronized estrus or ovulation for timed artificial insemination (AI) (7). The growth in average herd size and a reduction in the number of workers per animal make visual detection of estrus challenging. This situation has driven the development of numerous heat detection aids, including painting/chalking the tailhead, rub-off or activation of tail-head patches, and automated estrus detection systems. The latter are gaining popularity as the cost, ease of use, and reliability of the technology improve and farm labor becomes scarcer and more expensive. However, due to physiological variability among cattle (8) and potential technology failures (9, 10), it is unlikely that estrus detection aids will be the sole method used to identify nonpregnant animals for reinsemination.

Numerous protocols for timed AI have been developed and used in both dairy and beef farms, such as the Ovsynch protocols (11–13), estradiol plus progesterone-based protocols (E2+P4) (14), and their many variants. Ovsynch-like protocols employ gonadotropin-releasing hormone and



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**Table 1 Reproductive characteristics of select livestock species[a]**

| Species | Age at puberty[b] (months) | Time of pregnancy diagnosis[c] (days) | Earliest pregnancy status diagnosis[d] (days) | Gestation length[e] (days) | Interval between births[f] (months) |
|---|---|---|---|---|---|
| Cattle | 9–12 | 28–45 | 21–30<br>blood/plasma test | 278–288 | 12–14 |
| Sheep | 5–7 | 30–40 | 25–30<br>blood/plasma test | 145–150 | 12 or 5–8 for 2 times/year breeding |
| Goats | 4–12 | 30–40<br>ultrasound | 25–30<br>blood/plasma test | 150–155 | 8–12 |
| Pigs | 5–8 | 27–35<br>ultrasound | 17–20<br>blood/plasma test | 114–116 | 4–5 |
| Horses | 12–15 | 14–16<br>ultrasound | 9–10<br>ultrasound | 320–365 | 11–12 |
| Camels | 36–48 | 14–16<br>behavioral<br>40–60<br>transdominal ultrasound | 14<br>blood/plasma test and ultrasound | 360–390 | 24–26 |
| Llamas | 10–12 | 14–16<br>behavioral<br>45–60<br>transdominal ultrasound | 14<br>blood/plasma test and ultrasound | 330–390 | 12–16 |
| Alpacas | 10–12 | 14–16<br>behavioral<br>45–60<br>transdominal ultrasound | 14<br>blood/plasma test and ultrasound | 320–390 | 12–16 |

[a]The values indicated represent species averages, and actual values may vary based on factors such as breed, nutritional status, health, and management practices.
[b]Age at puberty indicates when an animal achieves sexual maturity for reproduction.
[c]Time of pregnancy diagnosis is the common gestational period when pregnancy is detected.
[d]Earliest pregnancy status diagnosis denotes the earliest time possible for detection of pregnant or nonpregnant status using modern methods.
[e]Gestation length is the duration from conception to birth.
[f]Interval between births is the usual time between consecutive births for an individual animal.

prostaglandin F2 alpha ($PGF_{2\alpha}$) to synchronize follicular growth, luteal regression, and ovulation (11). On the other hand, E2+P4 protocols use estradiol and progesterone to control follicle growth, $PGF_{2\alpha}$ to induce luteal regression, and estradiol to induce ovulation (14). Regardless of the method used to induce ovulation in cows, insemination cycles continue until pregnancy is achieved or until it is no longer economically viable to pursue a pregnancy.

Cows that remain pregnant until 45 to 60 days before calving are removed from the milking herd and enter the dry period, during which milking ceases. Dry cows are typically separated from milking cows to provide them access to a special diet that helps optimize feed intake and support rapid fetal growth without causing excessive body condition (weight) gain prior to calving. Under optimal conditions, this sequence of events will lead to a calving interval of 12 to 14 months. Beef cattle are managed like dairy cattle, but with a target of a 12-month calving interval that aligns with the yearly pattern of forage availability and maximizes marketing opportunities.

**Sheep/goats.** Many commercial breeds of sheep and goats are seasonally polyestrous, being born in the spring and initiating cyclicity at 5 to 7 months of age during the late summer and early fall in the northern hemisphere (15). Producers manage these animals so they reach puberty and conceive



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in the fall, with lambing occurring after a 5-month gestation in the spring at approximately 1 year of age. After the breeding season, animals that did not become pregnant are typically sold to minimize the costs of infertility. Less seasonal breeds can be strategically managed to lamb or kid in the fall (or year-round). When combined with early weaning practices, these breeds can generate 2 lamb/kid crops within a 12-month period. Sheep or goats raised for milk production are often managed to lamb/kid to maximize use of the available forage. However, given the high value of sheep/goat milk and cheese, more intensive, year-round production systems that rely on purchased or stored feeds are also common.

**Swine.** Gilts from current commercial maternal lines reach puberty at 5 and 7 months of age (80–120 kg), which allows the first litter to be farrowed close to 1 year of age. Practices such as mixing gilts, transport-induced stress, and daily boar exposure can help advance the onset of puberty. Prepubertal gilts at least 5.5 months of age and mature gilts that do not express estrus within 28 days of weaning can be treated with a combination of equine chorionic gonadotropin (CG) and human CG (trade name PG 600) to induce estrus within 4 to 10 days (16). These work by inducing growth and ovulation of tertiary follicles. However, most gilts are inseminated as they come into natural estrus. Another common practice is to refrain from inseminating gilts during their first estrus to allow for continued growth, improvement in oocyte quality, and an increase in ovulation rate that typically occurs by the second estrus. On average, estrus can last 24 to 72 h (sows tend to express longer estrus of 72 to 96 h), with ovulation occurring approximately 30 to 40 h after the initiation of estrus.

In modern commercial swine production, few females are inseminated by using a boar (natural cover). Large breeding units rely almost exclusively on AI on the first and second day (with a third day, if necessary) of estrus. The timing of ovulation can be regulated in weaned sows by using triptorelin, a gonadotropin-releasing hormone agonist that stimulates the surge of luteinizing hormone (LH) between 40 and 48 h after treatment (17). This regulated ovulation allows for precise scheduling of insemination for single fixed-time AI. The estrous cycle in pigs ranges from 18 to 22 days (average of 21 days), with corpora lutea regression occurring shortly after day 15 of the estrous cycle and final preovulatory follicle growth taking 3 to 5 days before returning to estrus.

Sows are grouped-weaned to synchronize breeding after 21 to 24 days of lactation, aiming for a return to estrus 4 to 5 days following weaning. If that does not occur, weaned sows can be treated with PG 600 to stimulate estrus for AI (18). Farrowing rates in commercial swine herds are close to 90%, and a healthy, productive sow can successfully farrow 2.5 litters each year, with an overall production of 25 piglets/year (19). Top US swine farms aim to achieve over 30 piglets/sow/year. Improvement in litter size was first targeted by selection for increased ovulation rate (20). Since 2000, the average litter size has increased from 12 to 16 piglets due to the increase in ovulation rate, which currently averages over 25 oocytes/cycle in current commercial herds (19). Although the primary goal in the swine industry is to improve embryonic and fetal growth, the increase in litter size also brings an elevated risk of embryonic loss and a greater number of low-birth-weight piglets.

**Horses.** Horses are seasonally polyestrous and long-day breeders. However, 10–20% of mares cycle year-round. Although temperature and food availability can affect cyclicity, the breeding season is influenced primarily by the lengthening of daylight hours (photoperiod), coinciding with reduced melatonin secretion from the pineal gland. Breeding operations aim to produce one foal per year, made possible by the rapid postpartum uterine involution and resumption of ovarian cyclicity in mares. Normal conception rate can be achieved after breeding on the first postpartum estrus (foal heat) if ovulation happens at/over 10 days after foaling (21, 22). For some



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breeds, January 1 (in the northern hemisphere) and August 1 (in the southern hemisphere) serve as universal horse birthdays that are related to their seasonal cycle. This primarily stems from horse racing industry conventions to simplify age-related regulations in competitive racing or performance. Manipulation of seasonality and the establishment of pregnancy as early as possible in the breeding season are critical for marketing and training performance horses. The annual breeding cycle in horses consists of a breeding season (April to September), winter transition, anovulatory/anestrus period, and spring/vernal transition (23). To achieve foaling in January, mares and stallions undergo light and/or hormonal treatments to advance the breeding season to February 15. Mating systems in horses include pasture mating; hand mating; AI with fresh, chilled, or frozen semen; and transfer of in vivo– or in vitro–produced embryos (24). For thoroughbred mares registered with the Jockey Club, natural mating is the sole breeding method permitted for foal registration.

**Camelids.** Camels are seasonal breeders, with the reproductive season aligning with shorter days, lower ambient temperatures, and increased rainfall. Three categories of breeding management systems in camels depend on whether animals are raised in a nomadic setting, for intensive dairy production, or as performance animals (racing, show). In nomadic system, males are grouped with females at a male:female ratio of 1:15 to 1:50 during the breeding season. In intensive dairy production and for performance animals, females are mated in small groups or in-hand mated to ensure accuracy of breeding dates. In camels, lactation significantly delays postpartum return to cyclicity. The general reproductive goal in this species is to achieve one parturition every 18 to 24 months (25, 26). In South America, the prevailing breeding management system for alpacas and llamas involves seasonal pasture breeding, typically spanning 60 to 90 days during the rainy season. In this system, males are grouped with females at a male:female ratio of 1:20 to 1:40, with males resting or changing groups every 1 to 2 weeks. In other geographic locations, such as the United States, Australia, and Europe, alpacas and llamas are bred year-round, either as small groups or by in-hand mating. In recent years, most owners have restricted breeding to the fall or spring to take advantage of the best environmental conditions. Rapid postpartum uterine involution enables the resumption of cyclicity approximately 10 to 15 days after parturition. The overarching reproductive objective is to achieve one offspring per year (27, 28).

## The Physiology of Early Pregnancy Provides Potential Options for Pregnancy Diagnosis

During pregnancy, intrauterine hormonal and physical signals coordinate maternal endocrine responses that induce dramatic changes in the reproductive tract, supporting maintenance of pregnancy and development to term. Some of the most critical chemical/hormonal signals, especially in early days of pregnancy, involve secretion of conceptus-specific proteins or other molecules, some of which are vital for maintaining the corpus luteum (CL). Interestingly, not all species require conceptus signals to maintain function of the CL throughout the length of gestation. For example, following a sterile mating, cats and dogs will form and maintain the CL for roughly the length of their gestation.

**Cattle.** The length of the estrous cycle in cattle ranges from 18 to 24 days. Cows spontaneously ovulate after estrus at approximately 24–28 h after an LH surge. The CL is fully developed by day 6–7 after ovulation, and without fertilization, luteolysis occurs between days 17 and 20 of the estrous cycle. The embryo enters the uterus at the morula stage approximately 4 to 5 days after ovulation. Development progresses to the blastocyst stage in the uterine lumen, with subsequent hatching from the zona pellucida occurring between days 8 and 9 after ovulation (29). Changes in endometrial gene expression become evident at the blastocyst stage of pregnancy (30, 31).



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Moreover, conceptus growth after hatching leads to increased production of various molecules by the rapidly elongating placental trophectoderm. It is essential for these events to occur before the onset of luteal regression (32).

The primary conceptus signal that is necessary and sufficient to rescue the CL is an immune cytokine, interferon tau (IFNT; 33). Produced by the conceptus trophectoderm in both glycosylated and non-glycosylated isoforms, IFNT secretion begins around day 10 and continues until approximately day 35 to 40 of gestation (33). IFNT signaling triggers transcription of several hundred interferon stimulated genes (ISGs) in the reproductive tract and peripheral tissues, including circulating immune cells around days 15 to 22 of pregnancy in cattle (34, 35). One consequence of this signaling is altered $PGF_{2\alpha}$ release by the endometrium that leads to the rescue of the CL. Therefore, IFNT, ISG, and plasma concentrations of $PGF_{2\alpha}$ and progesterone are potential indicators of early pregnancy in cattle.

During rapid conceptus elongation between days 13 and 20 of pregnancy, the chorion attaches to the endometrial epithelium. A defining feature at this time is the appearance of a population of giant binucleated cells (BNCs; 36). These cells produce, store, and secrete multiple kinds of signaling molecules, including a superfamily of glycoproteins encoded by several genes, known as pregnancy-associated glycoproteins (PAGs; 37–39). During placenta formation and throughout gestation, BNCs migrate into the endometrium and fuse with endometrial epithelial cells, forming large, multinucleated syncytial plaques (40). In response to cell fusion and/or migration, PAGs are released from cytoplasmic granules of BNC into the endometrium and adjacent to blood vessels, ultimately reaching maternal circulation (40). Although the biological function of PAG is currently poorly understood (41), detection and estimation of concentrations of PAG in maternal blood or milk are used as a method for pregnancy diagnosis because PAGs are produced only by placental cells (38, 40–42). Within the first trimester of gestation in cattle, the placenta begins to produce progesterone as well as other steroid hormones.

**Sheep/goats.** The estrous cycles of sheep and goats average approximately 16 and 19 days, respectively, with some variations depending on the breed. These species ovulate toward the end of estrus, roughly 30 h after the LH surge and fertilization occur in the oviduct. The embryo arrives in the uterus around day 4–5 at the morula stage and develops into blastocyst and hatches by day 9. Hatching is followed by rapid elongation of the conceptus trophectoderm starting at day 11–12. By day 17, the conceptus has elongated well into the opposite (of the ovulated follicle) uterine horn. Like those of cattle, the conceptus of sheep and goats produces IFNT, which initiates maintenance of the CL (43). In sheep, IFNT production begins around day 10 and terminates by day 25 of pregnancy. Interestingly, in contrast to that of cattle, ovine IFNT is not glycosylated. Nonetheless, it also induces the transcription of hundreds of ISGs in the endometrium and peripheral tissues (44, 45), providing opportunities to assay a similar array of endocrine responses to pregnancy. These ISGs are detectable in the peripheral blood starting around days 13 to 15 of pregnancy in sheep (34). Between days 15 and 17, the conceptus begins a more intimate attachment to the uterine wall. This is accompanied by the appearance of BNCs and production of PAGs, as described for cattle (36). By day 55 of gestation, production of progesterone by the placenta is sufficient to supplant that produced by the CL.

**Swine.** The estrous cycle of the pig averages 21 days, with ovulation occurring toward the end of estrus. Modern commercial swine will ovulate 15–25 oocytes, which are fertilized in the oviduct. Embryos reach the tips of the uterine horns on day 5 of gestation at the morula stage, where they develop into blastocysts by day 6 to 7. Subsequently, blastocyst growth and the production of blastocyst/endometrial enzymes lead to stretching of the zona pellucida and blastocyst hatching on day 8 (46).



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From day 8 to 12 of gestation, the spherical conceptuses are propelled by contractions of the uterine musculature between the uterine horns to establish equidistant spacing among each other. These contractions occur in response to histamine, estrogens, and prostaglandins secreted by the conceptuses (47). This movement ends after rapid elongation of conceptuses (occurring over 1 h) from spherical (10 mm) into filamentous (~150 mm) shape. Morphological changes during elongation are triggered by conceptus's production of interleukin-1β2 (46). Interleukin-1β2 also activates nuclear factor κβ signaling in the luminal epithelium of the endometrium underlying the attaching conceptuses. The signaling via nuclear factor κβ is important to regulate immune responses to conceptus attachment. Moreover, conceptus's secretion of estrogens and prostaglandins helps prepare the endometrium for pregnancy (48).

The process of conceptus elongation in pigs is crucial for the formation of adequate placental surface areas for each conceptus. Elongation is also critical to pregnancy establishment, as the pig requires at least two embryos developing within each uterine horn to inhibit synthesis and release of $PGF_{2\alpha}$ and to prevent luteolysis (49). Following elongation, attachment, and differentiation of the placental membranes, the pig conceptuses continue to secrete prostaglandins and exhibit a second rise in estrogen production from day 15 to 30 of gestation. Additionally, conceptuses also secrete interferon delta and interferon gamma, which are type 1 and 2 interferons, respectively (46). Therefore, estrogens, interleukin-1β2, prostaglandins, and interferon gamma, all critical for conceptus development and survival, could provide markers of early pregnancy in the pig.

The pig conceptuses and placenta also secrete PAGs (50). However, PAGs are not detectable in circulation of pregnant pigs, most likely due to lack of trophoblast BNC formation and fusion with the endometrial luminal epithelium, as observed in ruminants. After formation of the epitheliochorial placenta between days 20 and 35 of pregnancy, the allantois differentiates and expands, accumulating fluid. Ultimately, rapid accumulation of fluid in the allantois provides opportunity for detection of pregnancy after day 24 of gestation using ultrasonography (51, 52).

**Horses.** The estrous cycle spans 18 to 24 days, with the estrus phase lasting 4 to 8 days. Ovulation typically takes place in the final 1 to 2 days of estrus, after a prolonged elevation of LH. In nonpregnant animals, the CL lyse in response to endometrial secretion of $PGF_{2\alpha}$ around day 15 of the estrous cycle (53). However, the mechanisms that inhibit luteolysis remain elusive but likely involve both mechanical and chemical signals (54). After fertilization, only a viable embryo can pass through the uterotubal junction, facilitated by conceptus secretion of PGE2, which relaxes the uterine papilla, and $PGF_{2\alpha}$, which contracts oviductal muscularis (55). The embryo arrives at the uterus around day 5 to 6, as either a morula or early blastocyst. Subsequently, uterine contractions propel the blastocyst throughout the uterine horns and body, resulting in approximately 12 transits per day (56). This transuterine embryo movement is important for maternal recognition of pregnancy (MRP), because it ensures extensive contact of the embryo with the endometrium, preventing systemic release of $PGF_{2\alpha}$ (55, 57). In fact, termination of pregnancy can occur due to the presence of endometrial cysts blocking conceptus movement, or through uterine horn ligation (58, 59). Regulation of embryo movement likely involves PGE2 produced by the conceptus or by the endometrium in response to the embryo (60, 61).

Around day 7, the trophectoderm secretes an acellular capsule that replaces the zona pellucida. This spherical capsule facilitates transuterine embryo movement from day 7 to 16, by containing the growing blastocyst and providing mechanical protection and anti-adhesive properties due to its sialic acid content (61). Three key proteins in the capsule, uterocalin, insulin-like growth factor binding protein 3, and heparin-binding epidermal growth factor, are considered important for pregnancy establishment (62). Interestingly, the lifespan of the CL can be prolonged in mares via placement of glass marbles or self-assembling magnets in the uterine lumen, likely due to their



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mechanical effects on endometrial prostaglandin production. The role of the capsule in embryo fixation to the uterine wall after day 16 is not understood fully, but fixation involves changes in the capsule proteins, including changes in protein distribution (uterocalin), limited proteolysis (β2-microglobulin and uterocalin), and protein abundance (phospholipase A2) (63, 64). Moreover, fixation is also facilitated by an increase in conceptus size, elevated uterine tone, and reduced sialic acid content of the capsule (55).

The capsule is shed by day 23, permitting trophectoderm–endometrial interaction for implantation (61, 65). From days 35 to 40, chorionic cells interdigitate with endometrial epithelium, forming a microcotyledonary epitheliochorial placenta. Binucleated cells form from a specialized band of cells called the chorionic girdle and invade the endometrium, forming cuplike structures (66). Endometrial cups are present for 60 to 100 days and secrete CG, which is responsible for ovulation and/or luteinization of follicles, resulting in the formation of accessory CL and inducing a further increase in serum progesterone (66). Maintenance of the endometrial cups involves immunomodulation and split immune tolerance, with invasive trophoblasts not expressing major histocompatibility class II (MHC II) molecules, eliciting an immune response. Paternal MHC I are highly expressed on the chorionic girdle starting from day 30, gradually decreasing by day 45 of gestation. Robust but nonharmful production of maternal anti-fetal/paternal MHC I antibody starts at day 30 to 60 and increases in subsequent pregnancies due to immunological memory (67).

Endometrial cups and chorionic girdle cells produce pregnancy-specific glycoproteins, with endometrial cups possibly undergoing hypovascular (hypoxic) apoptosis (66). Production of placental progestogens starts around day 70, with 5α-dihydroprogesterone being the major progestin. In addition, fetal gonads release dehydroepiandrosterone and pregnenolone that are converted to estrone and 5α-pregnane in the allantochorion. These compounds are further metabolized into estrone sulfate in the endometrium and are detectable in maternal blood (68).

**Camelids.** As induced ovulators, camelids experience an LH surge triggered by β-nerve growth factor in seminal plasma, leading to ovulation approximately 30 h after mating (69, 70). The CL fully develops by day 8 to 9 and, in the absence of fertilization, luteolysis occurs by day 10 or 11 in response to $PGF_{2\alpha}$ pulses. Nonpregnant females typically return to estrus 14 days (approximately 2 weeks) after mating, whereas in pregnant females, luteolysis is prevented, and the CL remains the sole source of progesterone throughout gestation (26, 71, 72). Selective transport through the uterotubal junction, as seen in horses, is not evident in camelids, and unfertilized oocytes degenerate or are eliminated in the uterus (73). The embryo develops into the hatched blastocyst stage within the oviduct in approximately 6 to 7 days, reaching the uterus around day 7 (26, 73, 74).

The hatched blastocyst rapidly expands and migrates to the left uterine horn by 8.5 to 9 days, where it starts to elongate and occupy the entire uterine cavity by day 12. In alpacas, 83% of embryos resulting from a right ovary ovulation were in the left uterine horn on day 9 (73), and by day 15, the trophectoderm was in contact with the entire endometrial surface, though not yet attached. Despite both ovaries being equally active, embryo migration to the left uterine horn is essential for maintenance of pregnancy (73, 74). In the dromedary, higher pregnancy rates were also observed when embryos were transferred to the ipsilateral uterine horn after right ovary ovulation, as opposed to left ovary ovulation (75).

The mechanism of MRP in camelids remains elusive (74, 76). Current studies suggest this process likely happens between days 8 and 10 after mating to prevent luteolysis. Despite the conceptus's presence, pulsatile release of $PGF_{2a}$ occurs between days 7 and 15, causing a temporary dip in progesterone concentrations by day 9 (77, 78). Moreover, oxytocin plays no role in regulating luteolysis or maintaining pregnancy in camelids (79), and substances analogous to IFNT have not been identified (80). Instead, the period of MRP coincides with rapid conceptus elongation and



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a decrease in cyclooxygenase-2 expression in the endometrial luminal epithelium (81). Moreover, camelid conceptuses exhibit heightened aromatase (CYP19A1) activity between days 10 and 15, leading to increased estradiol 17β synthesis (82, 83).

Estrogens' involvement in MRP is supported by elevated estrogen receptor α (ESR1) between days 8 and 12 after mating in pregnant animals and a reduction in progesterone receptor expression by day 12 (81). In llamas, no differences in endometrial expression of ESR1 and ESR2 mRNA were observed between pregnant and nonpregnant individuals on days 7, 9, and 11 after mating (84). However, pregnant llamas exhibited intensified ESR1 expression in stromal cells on day 8 and in the endometrial luminal epithelium on day 12 after mating, suggesting paracrine effects of blastocyst-produced estrogen in inducing localized endometrial changes for MRP (83, 84). By day 12, decreased progesterone receptor in luminal and glandular endometrial epithelia was noted in pregnant llamas, potentially influencing gene expression related to histotrophe production, embryo attachment, and placentation (85). Additionally, pregnant alpacas experienced a decline in mucin 1 and mucin 16 around the same period, implicating their role in preparing the endometrium for placentation (86).

Embryo attachment starts around day 14 or 15 of pregnancy (87, 88). During the precontact phase, the mononuclear trophoblast cells are separated from the endometrium by histotrophe. Then, well-formed microvillus junctions develop gradually between fetal and maternal tissues to establish the inter-areolar regions by day 45 in alpacas (87) and day 56 to 70 in the dromedaries (82, 89). Giant multinucleated trophoblastic cells form because of either fusion of mononucleated trophoblasts (87) or mitotic polyploidization (90). Camelid placentation is epitheliochorial, microcotyledonary, and diffuse. Moreover, differences like uteroplacental protein glycosylation may explain the difficulty in establishing interspecies pregnancies between camels and new-world camelids (91). To date, attempts to detect PAGs for pregnancy diagnosis have been unsuccessful. PAGs, found in other ruminants, were not detected in llama sera between 2 and 244 days of pregnancy (92). Placental extracts from alpacas and dromedaries showed no immunoreactivity with PAG-1 family molecules but did react with antiboPAG-II serum (93). However, immunolocalization revealed chorionic PAGs in alpacas, named LpPAG (94). Similarly, in *Camelus dromedarius* and *Camelus bactrianus*, PAGs with a structure like that found in pigs were identified within trophectoderm cells and named CdPAG and CbPAG, respectively (95).

## Current Methods for Pregnancy Detection in Livestock

Pregnancy detection methods can be grouped broadly into indirect and direct categories. Indirect methods assess cyclic status or confirm the presence/absence of progesterone after breeding. Visual detection of estrus after insemination can be ineffective due to undernourishment-induced anestrus or pregnancy loss or in situations where females do not exhibit signs of estrus in the absence of males. Identification of failed pregnancies through low progesterone after insemination at a time that coincides with the predicted onset of the next estrus is highly accurate. However, because this method requires precise sampling intervals, in large herds and/or understaffed farms, blood sampling may not be practical and can generate false-positive results because cyclic animals in the luteal phase also present high circulating concentrations of progesterone.

Direct methods for pregnancy detection offer higher accuracy by physically identifying the conceptus (palpation or ultrasound) or measuring a product produced by the conceptus (pregnancy-specific molecules). In cattle and horses, TRP of uterine contents can be highly accurate and cost effective. On the other hand, transrectal real-time (B-mode) ultrasonography (TRUS) is an effective and widely used method for pregnancy testing (7) that allows practitioners to visualize the conceptus(es), assess viability, measure size, determine sex, and evaluate ovarian



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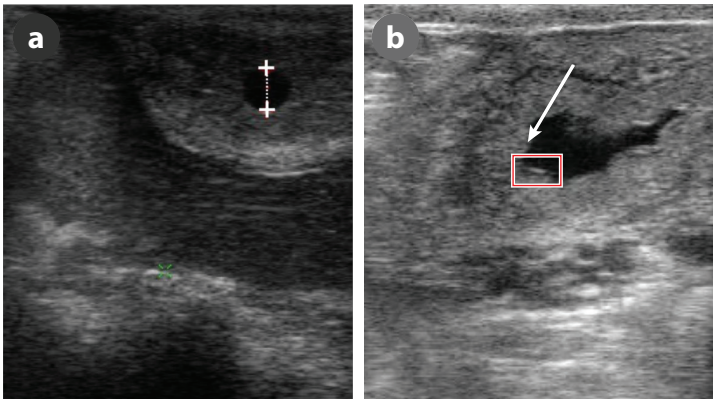


**Figure 1**

Transrectal ultrasonography images of (*a*) a day-10 uterus of a mare with a capsulated embryo of 6.7 mm diameter (*plus signs*) and (*b*) a 27-day fluid containing the uterus of a cow with an elongating embryo (*arrow*).

structures. Ultrasound technology converts electrical signals into ultrasound waves using piezoelectric crystals housed in the probe. The sound waves are variably reflected depending on tissue density and then converted by the probe into electrical signals hundreds of times per second that generate a real-time image on a screen. Tissues with the greatest density (e.g., bone and cartilage) reflect more sound waves and appear white, whereas less dense tissues will be gray, and fluid-filled spaces that reflect few sound waves appear dark or black (**Figure 1**). TRUS also requires training, and equipment costs can be prohibitive. Contract services for TRP and TRUS can be expensive and often do not allow for optimal timing of pregnancy diagnosis on farms where scheduled visits of veterinary practitioners occur infrequently (7).

**Cattle.** Both TRP and TRUS are used routinely to identify nonpregnant cows for rapid reinsemination (7). Although it is possible for some practitioners to conduct the procedure earlier under research settings or when time is not a constraint, TRP is commonly performed around 35 days of gestation or later (3). For TRUS, pregnancy can be detected in research settings as early as day 20 to 22 after insemination in heifers (96, 97). However, studies in commercial conditions conducted with many animals, or with lower-resolution equipment, reported acceptable performance after 25 days for heifers and 28 days for cows (98, 99). TRUS offers the advantage of evaluating ovarian structures during pregnancy diagnosis (100, 101). Determination of the presence and size of corpora lutea, follicles, and cystic ovarian structures enables targeting subgroups of cows for specialized fertility protocols (13, 102, 103). This improves reproductive performance and reduces costs in dairy herds (104). Another benefit is fetal sexing, which is possible starting at approximately 50 days of gestation and has an optimal window between 55 and 80 days, depending on the cow's size (105, 106).

Chemical tests based on the detection and estimation of concentrations of hormones or other molecules are also available for cattle. Secreted by placental cells, concentrations of PAG in blood or milk can accurately diagnose pregnancy starting around 25 and 35 days after insemination, respectively (107–109). PAG can be detected in circulation of pregnant cows as early as day 15 (38, 42). Nevertheless, due to biological variability in the timing and magnitude of the initial rise in PAG secretion, commercial detection of this molecule is generally recommended for blood samples of cattle after day 25 of gestation (38, 42, 110). Furthermore, commercial assays can be used to detect PAG in the blood not only of pregnant cattle but also of other species, such as



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sheep, goats, bison, deer, and elk. Timing of sampling can be arranged to enable implementation of effective resynchronization programs upon arrival of test results, allowing open cattle to receive two inseminations in 35 days (107). Beyond some potential limitations associated with sampling, shipping, and on-site testing, PAG-based pregnancy tests can also generate false-positive results for cows tested in early lactation (111–113) and cows with a recent pregnancy loss (109, 113). This is due to the slow rate of decline of PAG in circulation, resulting in PAG from a prior pregnancy persisting well into the next lactation or for days to weeks after a pregnancy loss (42, 113, 114).

**Sheep/goats.** Most sheep and goats are bred by rams/bucks with relatively little use of AI or embryo transfer. Generally, females are housed with males for 30 to 45 days or through the entire breeding season. Pregnancy diagnosis is accomplished 4 to 6 weeks after the male has been removed. Mating progress can be monitored by fitting the male with a marking harness or painting the brisket of males with a colored grease. The colors are changed every two weeks (lighter to darker colors) to identify females that were mated and then returned to estrus. Animals that are not marked by the end of the breeding season can be culled, along with those identified as nonpregnant 4 to 6 weeks after the male is removed. Pregnancy diagnosis can be accomplished via transrectal or transabdominal ultrasound (TAUS) as early as day 25 to 30 or 35 to 40 of gestation, respectively. Furthermore, commercial assays for PAG are also available for sheep and goats as early as day 30 of gestation (115). Pregnancy diagnosis over 90 days of pregnancy can be accomplished using the ballottement procedure.

**Swine.** In the swine industry, pregnancy detection involves observing behavioral changes associated with estrus return and using ultrasonography to monitor embryo presence. Absence of conceptuses before day 15 after insemination leads to CL regression between days 15 and 17. Nonpregnant females experience accelerated follicle growth and increased estrogen production and exhibit vulva swelling, redness, heightened activity, and vocalization. Detection of return to estrus, often around day 17, involves monitoring standing estrus behavior (immobility), with or without a boar's presence. Signs include immobility, erect ears, and ears flipping against the neck. Boar assistance enhances estrus detection.

When estrus is not detected after insemination, females undergo ultrasound for confirmation of pregnancy (51, 52). The accuracy of this method depends upon the type and day of detection. Amplitude depth (A-mode) ultrasonography has the advantage of being simple, inexpensive, and easy to use. However, A-mode ultrasonography is not accurate until after day 28 of pregnancy, when there is a large amount of fluid in the allantois. In addition, the technician must be careful to position the probe correctly on the flank, especially in earlier days of pregnancy, to avoid false positives through detection of fluid in the bladder. Doppler ultrasonography for detection of increased blood flow or a fetal heartbeat is accurate when used after day 29 of pregnancy but requires the technician to have a high degree of training. Although a more expensive option, real-time (B-mode) ultrasonography offers precise pregnancy detection in swine as early as day 21 to 23 of gestation. Most farms perform this type of detection once weekly when sows reach 24 to 30 days after insemination. Early imaging on day 19 is possible but less accurate due to minimal allantoic fluid and potential false positives caused by estrus-related uterine fluid accumulation.

Hormonal tests can also be employed for monitoring pregnancy in pigs. Low progesterone concentration in the blood after day 15 of gestation indicates nonpregnant females before their return to estrus. An available commercial estrone sulfate test kit can detect pregnancy after day 23, as the endometrium sulfates conceptus and placental estrone before release into maternal blood. However, the requirement for sample collection (blood, urine, or feces) and the time-consuming, on-farm analysis have limited the widespread use of this method in monitoring pregnancy status.



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**Horses.** Traditionally, pregnancy diagnosis was based on nonreturn to estrus after mating or insemination. However, in modern equine breeding, TRUS is the most common pregnancy diagnostic approach. TRUS can detect the embryonic vesicle as early as 9 to 10 days after ovulation, but in practice, TRUS is performed at 14 days after breeding with 100% accuracy, being the earliest pregnancy diagnosis in livestock. Detection of pregnancy on day 14 is important for allowing prediction of the next estrus in nonpregnant mares and the elimination of one embryonic vesicle in cases of twin pregnancies. A second examination is performed on day 25 and 28 of gestation to confirm embryo heartbeat. Subsequently, viability of pregnancy is checked at day 60 of gestation, as most embryo loss in mares occurs before day 45. Fetal sex determination is possible between 60 and 80 days by transrectal examination and between 100 and 150 days by transabdominal examination. A late confirmation of pregnancy by ultrasonography is often performed in valuable mares to evaluate fetal well-being and placental integrity. Overall, the high cost of ultrasound examination in horses is justified by the greater value of their offspring.

Serum CG concentration can be used as pregnancy diagnosis between days 45 and 90 of gestation in mares. However, false positives may occur due to the persistence of endometrial cups after pregnancy loss. Accurate pregnancy diagnosis between days 80 and 100 can be achieved with serum, urinary, or fecal estrone sulfate measurements. Nonetheless, other available methods significantly reduce the value of these tests for domestic equids.

**Camelids.** Camels display a distinctive stance during pregnancy. When approached by a male, females stand still with an elevated head, slightly widespread hind legs, and their tail coiled upward. Observation of this behavior is the most common method for pregnancy diagnosis, with an accuracy of 60% to 94% (116). In alpacas and llamas, the most common behavior for pregnancy detection is "spitting off the male," reflecting aggressive behavior of the female toward the male (26). Serum progesterone above 2 ng/mL indicates a functional CL and a method for detection of pregnancy 14 to 16 days after mating. This method is accurate only if a breeding date is known but can yield false-positive results if a CL persists in the absence of pregnancy. Serum and urine estrone sulfate concentrations increase during pregnancy and peak at 21 days and in the last month of pregnancy (117).

Although TRP is a viable method for pregnancy diagnosis after day 40 of gestation (118), it is employed infrequently. TRUS remains the gold-standard method for pregnancy diagnosis in camelids because it can be performed as early as 18 days after mating with 100% accuracy. With TRUS, pregnancy is confirmed by visualization of the embryonic vesicle and the CL at day 18 and a fetal heartbeat by day 25 after mating. In alpacas and llamas, pregnancy diagnosis by percutaneous transabdominal ultrasonography can be performed with great accuracy after day 40. However, beyond day 100, pregnancy diagnosis by this method requires clipping the fiber of the abdominal region for better visualization of the uterus and fetus (119, 120).

## Emerging Methods for Pregnancy Detection in Livestock Species

The development of new technologies for livestock pregnancy diagnosis is driven primarily by the costs associated with open days in nonpregnant animals, overall fertility in the flock/herd, the value of successful pregnancies, economic losses from culling, and the high labor costs on farms. These considerations typically are more pronounced in horses, swine, cattle, and camelids. Consequently, the adoption of new methods, especially those employing computer-aided, passive, and noninvasive technologies leveraging artificial intelligence, is expected to increase in intensive production systems. However, the economic incentives for developing such technologies are relatively low for small ruminants, as explained below.

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**Cattle.** Doppler TRUS is gaining popularity as a method of identifying nonpregnancy (121, 122). Based on the negative correlation between the area of the CL with blood flow and circulating concentrations of P4 (123), it is possible to determine functionality scores of CL based on its blood flow visualization with Doppler TRUS. Initial reports showed 74.8% accuracy of this method in diagnosing pregnancy 20 days after insemination in dairy heifers and cows (121). When identifying nonpregnant cows, Doppler TRUS showed lower specificity (53.7%) and positive predictive value (65.1%). This discrepancy is attributed to a high proportion of cows with a functional CL around 20 days after AI, later confirmed to be not pregnant. Recent studies on Doppler TRUS in dairy (122) and beef (124, 125) cattle have reported similar results, highlighting its applicability in reproductive management programs, including AI (125, 126) and embryo transfer (127, 128). Challenges to the widespread adoption of this method include the additional cost of equipment; the need for extra training and time for accurate implementation in commercial settings; and the necessity to confirm pregnancy after the initial examination due to inconclusive positive results, particularly in adult dairy cattle.

In-line milk sampling offers opportunities for developing pregnancy testing technologies using reproductive status markers or physicochemical changes in milk. For instance, automated methods that measure hormone concentrations, like Herd Navigator™ from DeLaval Inc., have been commercially available for more than a decade. These systems employ automated biosensor technology to sample and quantify milk progesterone concentrations at preset intervals (129–131). Depending on the sampling frequency chosen, this method can be effective at determining return to estrus after insemination and for pregnancy confirmation after day 40 of pregnancy (129, 130). Furthermore, some studies suggested that pregnancy success could be predicted as early as 10 to 15 days after insemination based on elevated milk progesterone concentrations (131). Despite its accuracy for commercial farms, the high cost, lack of interoperability with other milking systems, and regulatory challenges in some countries have limited the adoption of this technology. Another emerging in-line testing method for dairy cattle pregnancy is Fourier transform infrared spectroscopy of milk. Although current studies show limited accuracy and commercial value (132–134), technological advances may address accuracy concerns. However, a current limitation is the unavailability of equipment for in-line testing on commercial farms.

Conceptus-produced IFNT induces expression of ISG broadly throughout the body during early pregnancy in ruminants. Yankey et al. (45) were the first to demonstrate an increase in ISG in peripheral blood leukocytes during early pregnancy in sheep. This effect of early pregnancy on expression of ISGs such as *MX1* and *MX2* in blood leukocytes was later confirmed in cattle (34, 35). Subsequently, IFNT-induced ISG expression was detected in the CL, liver, spleen, thymus, and bone marrow during early pregnancy in several ruminant species (35). This induction of ISG in the peripheral tissue was the result of IFNT escaping the uterus (135). These discoveries have stimulated work on approaches for pregnancy diagnostics that measure ISG in extrauterine tissues/fluids (136, 137). The advantage of this method is that it would be possible to detect failed inseminations prior to the expected return to estrus (day 18 to 20 in cattle). This would facilitate resynchronization and reinsemination of nonpregnant cattle at 21-day intervals without estrus detection. More recently, Domingues et al. (138) showed that upregulated ISGs were detectable in cells sampled from the cervix of pregnant cattle, providing yet another sampling approach for new pregnancy tests. To date, no commercial diagnostic tests measure ISGs as markers of pregnancy status in ruminants.

**Sheep/goats.** Economic factors, such as high fecundability and low per-animal value, limit the development of new pregnancy diagnostic tools for small ruminants. TAUS is a cost-effective and efficient method for large-scale producers. Effective starting 30 days after insemination, TAUS

quickly identifies open animals for culling and accurately diagnoses multiple fetuses. Techniques developed for cattle, like PAG assays, are also applied to small ruminants. Additionally, low blood progesterone at 16 to 18 days after insemination effectively diagnoses failed conception, although it is used infrequently. Emerging technologies for interferon-stimulated genes, described above for cattle, could similarly be applied to sheep and goats due to their high production of IFNT during early pregnancy (33).

**Swine.** Emerging swine pregnancy diagnosis methods focus on estrus detection, using noninvasive technologies like video surveillance and computerized facial recognition (139). These technologies monitor sow activity, movement, and vulva swelling, reducing labor and estrus detection time. Traditional estrone sulfate testing between 20 and 25 days after insemination remains available (140, 141) for pregnancy diagnosis, and its application is expanding. For instance, it can be used to determine small litter size (140, 141). Thus, females could be culled from the farrowing group early to improve efficiency. However, this method has not been adopted by the industry, as producers use females with small litters to cross-foster piglets from females with litter sizes greater than the number of available nipples.

**Horses.** An emerging method for pregnancy detection in equines is the detection of early pregnancy signals, such as the immune-suppressive glycoprotein early pregnancy factor (EPF). EPF can be identified in maternal circulation just 2 days after fertilization (142), making it potentially beneficial for broodmare management, embryo collection decisions, and assessing factors affecting embryo survival. Also known as heat shock protein E 1, EPF is a member of the heat shock family of proteins that is detectable in blood in response to pregnancy and tissue damage (including ovulation) and in certain cancers. This molecule consists of an estrus-dependent component and a pregnancy-dependent [EPF(PD)] component (143, 144). EPF(PD) is produced in the ovary in response to a signal from the oocyte after fertilization has occurred (144). EPF can be detected using a rosette inhibition test that involves a complex laboratory assay not practical for routine clinical use. Moreover, diagnostic tests available for EPF detection for pregnancy diagnosis in mares remain unreliable (143, 145).

**Camelids.** Currently, the absence of a specific chemical signal for MRP and the delayed expression of PAG in camelids limit the adoption of hormonal methods for pregnancy detection. Until a distinctive early pregnancy marker is identified in camelids at 2 weeks of gestation, behavioral testing and transrectal ultrasonography will continue to be the most reliable methods for pregnancy diagnosis in controlled breeding. Furthermore, aside from its accuracy and early pregnancy detection, ultrasonography provides additional benefits, including the diagnosis of pregnancy-related disorders such as twinning and embryo/fetal loss.

## SUMMARY AND CONCLUSION

Early and accurate pregnancy diagnosis is fundamental to efficient and profitable livestock production. Methods and timing of diagnosis are driven by trade-offs between costs/ease of diagnosis, accuracy, and prevalence and timing of pregnancy loss. Numerous low-tech and inexpensive methods exist, but these are often inaccurate or can be used only later in gestation. High-tech methods, though they enable earlier and more accurate diagnosis, are often economically inaccessible for small-scale farmers, especially in low- and middle-income countries. These methods facilitate management of large herds/flocks and reduce labor costs. They also require skill and training of farm personnel for optimal use. Emerging technologies, incorporating noninvasive approaches (video surveillance, activity monitors, automated sampling) coupled with machine learning and artificial intelligence, hold the promise of significantly improving accuracy, ease of use, animal



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welfare, and the overall value of pregnancy diagnosis in livestock. In conclusion, the evolution and refinement of pregnancy diagnostic methods across various livestock species have played a crucial role in enhancing the economic efficiency of livestock production systems. The selection and application of these methods require a careful balance between accuracy, early detection, and cost-effectiveness.

## DISCLOSURE STATEMENT

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